

CHRONIX EDU

Product Requirements Document

Version 1.0 | Chronix Technology Limited

“ChroniX Edu is a school management system that helps Nigerian private schools automate student administration, manage academic results, track attendance, and connect teachers with parents — replacing spreadsheets and manual processes with a single, organised platform.”

Product Name	Chronix Edu
Owner	Chronix Technology Limited
Document Version	1.0
Status	ACTIVE — Pre-Development
Target Launch	September 7, 2025
First Client	Child Prime Onyx School
MVP Scope	Foundation + Phase 1 + Phase 2
Stretch Goal	Phase 3 (if timeline permits)

1. What We Are Building

Chronix Edu is a web-based, multi-tenant School Management System (SMS) built specifically for Nigerian private schools. It is a SaaS product — multiple schools subscribe and each gets their own isolated, configurable environment on shared infrastructure.

SAAS
PRODUCT

Chronix Edu is not a single-school application. It is a platform that can be sold to any Nigerian private school. Every architectural decision — database design, authentication, settings management — must support multiple schools running simultaneously on the same system.

1.1 The Problem We Are Solving

The vast majority of Nigerian private schools — including Child Prime Onyx School — manage their operations manually. This means:

- Student records kept in physical registers or spreadsheets
- Exam scores entered manually, positions calculated by hand
- Report cards typed or handwritten every term
- Attendance marked on paper, no historical record
- Parents receive information only through paper notes sent home with children
- No centralised system — information lives in different people's heads, files, and phones

This creates errors, delays, and a lack of visibility for school leadership. A principal cannot see in real time how their school is performing. A parent cannot check their child's attendance without calling the school. A bursar cannot quickly know who has not paid fees.

1.2 The Solution

Chronix Edu replaces all of this with a single, organised platform. The core product at launch covers:

Capability	What it replaces
Student Administration	Physical admission registers, manual record-keeping
Academic Results & Report Cards	Manual score calculation, handwritten or typed report cards
Attendance Tracking	Paper registers, no searchable history
Result Approval Workflow	Informal review process with no audit trail
Parent Communication	Paper notes, phone calls, WhatsApp messages
School Configuration	Developer changes for every school-specific setting
Principal Dashboard	No visibility — information gathered manually
Fee Management (Phase 3)	Manual receipt books, spreadsheet tracking

1.3 What Chronix Edu Is NOT

Scope boundaries are as important as scope. The following are explicitly out of scope for the September 7th launch:

- A learning management system (LMS) — no online lessons, video content, or e-learning
- A scheme of work or curriculum planning tool
- A payroll system for staff salaries
- An inventory or asset management system
- A real-time chat application (messaging is structured, not WhatsApp-style)
- A mobile app (Phase 4 — web app is mobile-responsive instead)
- AI features such as comment generation or performance prediction (Phase 4)

**SCOPE
RULE**

Any feature request that does not directly serve student administration, academic results, attendance, or parent communication is out of scope for launch. Document it, do not build it.

2. Who It Is For

Chronix Edu is built for Nigerian private schools — both primary and secondary. Within a school, it serves six distinct user roles. Each role has a different job to do and a different definition of value.

2.1 Target Market

Attribute	Primary Target	Secondary Target
School Type	Private Primary	Private Secondary
Location	Lagos & surrounding states	Other Nigerian states
Size	50–500 students	500–1,500 students
Current System	Manual / Excel	Basic software or Excel
Decision Maker	School Proprietor / Principal	School Proprietor / Board
Payment	Yearly subscription	Yearly subscription

2.2 First Client: Child Prime Onyx School

DESIGN TARGET	<i>Every design and build decision in Phase 1 and Phase 2 must be validated against Child Prime Onyx School's reality. If it would not work for this school, it is not ready.</i>
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School Name	Child Prime Onyx School
School Type	Private — Primary and Secondary
Location	Nigeria
Current System	Fully manual — no software in use
Migration Challenge	All historical data is on paper or spreadsheets
Launch Date	September 7, 2025 (new academic term)
Role in Product	Beta client and reference school for all early sales demos

2.3 User Roles

Six roles will exist in the system at launch. Each has a specific portal and permission set.

Role	Who They Are	Primary Job in the System
Super Admin	Chronix staff	Onboard schools, manage subscriptions, cross-school oversight
Principal	Head teacher / proprietor	Approve results, view school-wide data, manage settings
Teacher	Subject / class teacher	Enter scores, mark attendance, communicate with parents

Registrar	Admin / admissions staff	Register students, manage enrollment, generate transcripts
Parent	Student's guardian	View child's results, attendance, messages from school
Student	Enrolled student	View own results, assignments, school notices

3. What Done Looks Like

"Done" has three levels: features complete, school-ready, and launch-ready. A feature being coded is not done. Done means it works for Child Prime Onyx School on September 7th.

3.1 The September 7th Definition of Done

LAUNCH DEFINITION	<i>On September 7th, a teacher at Child Prime Onyx School must be able to sit down, log in, and use Chronix Edu without any developer assistance. If that is not true, the product is not done.</i>
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3.2 Core Feature Checklist — Foundation

These must be complete before any other feature can be built or tested.

Feature	Priority	Phase
Multi-tenant database schema with school_id isolation on all tables	CRITICAL	Foundation
Row Level Security — School A cannot access School B data	CRITICAL	Foundation
JWT authentication with role claims (school_id, role, user_id)	CRITICAL	Foundation
School onboarding — create school, upload logo, set colours	HIGH	Foundation
Academic session and term management with single-active constraint	CRITICAL	Foundation
Class and subject management with teacher assignment	HIGH	Foundation
Assessment configuration engine with 4-level resolution and weight validation	HIGH	Foundation
Admin settings panel — identity, academic structure, grading scale	HIGH	Foundation

3.3 Core Feature Checklist — Phase 1 (MVP)

Feature	Priority	Phase
Student registration with auto admission number and parent account creation	CRITICAL	Phase 1
Score entry per component with teacher assignment validation	CRITICAL	Phase 1
Result computation — weighted totals, grades, positions, at-risk	CRITICAL	Phase 1
Result approval workflow — DRAFT > SUBMITTED > APPROVED > PUBLISHED	CRITICAL	Phase 1
Report card PDF generation with school branding via Puppeteer	CRITICAL	Phase 1
Principal dashboard — school overview, result status, at-risk students	HIGH	Phase 1
Teacher dashboard — workload, score entry status, pending submissions	HIGH	Phase 1

Assessment config UI — weight builder with real-time validator	HIGH	Phase 1
User management UI — create, deactivate, assign teachers to classes	HIGH	Phase 1
Score entry UI — grid with live totals, bulk save, submission flow	CRITICAL	Phase 1
Principal result approval UI — colour-coded grid, two-step publish	CRITICAL	Phase 1
Settings locking — 423/400/409 guards and audit log on all settings changes	HIGH	Phase 1

3.4 Core Feature Checklist — Phase 2 (Launch Target)

Feature	Priority	Phase
Daily attendance marking with upsert (same-day correction allowed)	CRITICAL	Phase 2
3-consecutive-absent alert triggering parent notification	HIGH	Phase 2
Attendance calendar view per student — 30-day history	HIGH	Phase 2
Parent portal — multi-child, results, attendance, messages	CRITICAL	Phase 2
Student portal — results with term selector, notices	MEDIUM	Phase 2
Assignment module — create, submit, grade, deadline enforcement	MEDIUM	Phase 2
Behaviour and discipline tracking with automatic parent notification	HIGH	Phase 2
Messaging — structured inbox, announcements, notification worker	HIGH	Phase 2
Offline PWA — Dexie.js queue for attendance and score entry	HIGH	Phase 2
Registrar module — admissions, promotion manager, transcript	HIGH	Phase 2
Notification settings — SMS sender, event toggles, alert thresholds	MEDIUM	Phase 2

4. Build Timeline

13 weeks from start to September 7th. The timeline below is the plan. Any feature that is not complete by its phase deadline moves to post-launch, not to the next week. Scope is fixed. Time is fixed. Quality is non-negotiable.

RULE	<i>A half-built feature is worse than a missing feature. If attendance marking works but the calendar view is not ready, ship without the calendar view and add it in October. Do not ship broken.</i>
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Phase	Weeks	Key Deliverable	Duration	Status
Foundation	Weeks 1–3	Database, auth, RLS, school setup, session management	3 weeks	NOT STARTED
Phase 1 — MVP	Weeks 4–10	Students, scores, results workflow, report cards, dashboards	7 weeks	NOT STARTED
Phase 2	Weeks 11–13	Attendance, parent portal, messaging, offline PWA	3 weeks	NOT STARTED
Phase 3 (Stretch)	Post Sep 7	Fees, Paystack, analytics, timetable	TBD	STRETCH

PHASE 2 WARNING	<i>3 weeks for Phase 2 is tight. If Phase 1 runs over, Phase 2 scope must be cut — not the deadline. Parent portal and attendance are the non-negotiables. Assignments and behaviour module can be post-launch if needed.</i>
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5. Technology Stack (Locked)

The stack below is locked. No alternatives will be evaluated during the build. Evaluating alternatives during a 13-week build is a schedule killer.

Layer	Technology	Why
Frontend	Next.js 14 + Tailwind CSS	App Router, SSR, path aliases, PWA support
Backend API	Node.js + Express (TypeScript)	Separate service, clear API boundary
Database	PostgreSQL via Supabase	Relational data, RLS, complex aggregations
Auth	Supabase Auth + JWT	Custom claims (school_id, role) built-in
File Storage	Supabase Storage	Logos, photos, PDFs, report cards
PDF Generation	Puppeteer (headless Chrome)	Full CSS control for report cards
Real-time / Push	Firebase Cloud Messaging	Notifications only — not DB replacement
SMS	Termii	Nigerian gateway, per-school API key
Email	SendGrid	Transactional email delivery
Payments	Paystack (Phase 3 only)	Best Nigerian payment gateway
Offline Storage	Dexie.js (IndexedDB)	Queue attendance and scores offline
Hosting — Frontend	Vercel	Next.js native, global CDN
Hosting — Backend	Railway	Node.js/Express deployment
Monorepo	npm workspaces	/apps/web + /apps/api
Language	TypeScript throughout	Type safety across frontend and backend

6. Success Criteria

The product is successful when Child Prime Onyx School uses it as their primary system from September 7th and does not revert to spreadsheets. The following criteria define this in measurable terms.

6.1 Launch Success (September 7th)

Criterion	Measurement
All students at Child Prime Onyx School registered in the system	<i>Student count in DB matches school register</i>
Teachers can log in and mark attendance on day one	<i>Zero support calls for login issues on launch day</i>
Score entry works for all subjects and classes	<i>Teachers successfully enter and submit scores for Term 1</i>
Principal can approve and publish results	<i>Results published without developer intervention</i>
Report cards generated with school branding	<i>PDF output matches school's existing report card format</i>
Parents can view their child's results online	<i>Parent portal accessible on mobile at 375px</i>
No cross-tenant data leaks	<i>Automated RLS test suite passes 100%</i>
System available 99% of the time in the first month	<i>Vercel + Railway uptime monitoring confirms SLA</i>

6.2 What Failure Looks Like

These are the failure scenarios that must be prevented:

- A teacher at Child Prime Onyx School reverts to paper registers because the system is too slow or confusing
- A parent from School A can view a student from School B (data isolation failure)
- Results are published to parents before the principal approves them
- A score is changed after results are published with no audit trail
- The system is down on the first day of term
- Report cards are generated with the wrong school name or logo
- A developer is required to change any school setting (grading scale, logo, class names)

6.3 Non-Functional Requirements

Performance	All dashboard API endpoints respond within 2 seconds at 50 concurrent users
Mobile	Parent portal and teacher attendance UI fully usable at 375px screen width

Offline	Attendance marking and score entry work without internet connection (Dexie.js queue)
Security	All API routes require valid JWT. RLS enforced at database level. No route accessible without authentication.
Audit	Every score change, settings change, and result status change is logged with user, timestamp, old value, and new value
Uptime	99% availability minimum. Downtime during school hours (7am–5pm WAT) requires immediate escalation.
Data Privacy	NDPR compliant. Parent consent obtained at registration. Data Processing Agreement included in school contract.
Branding	Each school's portal shows their own logo, name, and colours. Chronix Edu branding is minimal and secondary.

7. Open Decisions & Risks

These are questions that must be answered before or during development. An unanswered decision becomes a wrong assumption in code.

7.1 Decisions Required Before Build Starts

Decision	Why It Matters	Status
How many terms per year for Child Prime Onyx School?	<i>Session/term architecture is built first — wrong count affects every data record</i>	CONFIRM WITH SCHOOL
What is the grading scale? (A=70-100, B=60-69, etc.)	<i>Default grading scale must be set at onboarding — wrong scale produces wrong grades</i>	CONFIRM WITH SCHOOL
What assessment components does the school use? (CA1, CA2, Exam weights)	<i>Assessment config engine resolution depends on this being set before score entry</i>	CONFIRM WITH SCHOOL
What is the yearly subscription price?	<i>Needed for the school contract and Paystack setup in Phase 3</i>	DECIDE: Chronix
What domain will the system run on?	<i>Vercel deployment requires a domain configured before Child Prime Onyx goes live</i>	DECIDE: Chronix
Will Termii SMS be enabled from day one?	<i>Affects notification worker setup and requires Termii API key per school</i>	DECIDE: Chronix

7.2 Top Risks

Risk	Level	Impact	Mitigation
Phase 2 scope too large for 3 weeks	HIGH	<i>Incomplete parent portal or offline PWA at launch</i>	Cut assignments and behaviour module first — ship parent portal and attendance only
School data model wrong before code is written	CRITICAL	<i>Schema changes after data exists cause expensive rework</i>	ERD must be reviewed and signed off before Foundation build starts
Child Prime Onyx School not ready to migrate on Sep 7	MEDIUM	<i>Delay in data entry means teachers can't use system on day one</i>	Begin student data import 2 weeks before launch
AI coding assistant generates inconsistent code style	MEDIUM	<i>Codebase becomes unmaintainable across 47 prompts</i>	Agent file (rules document) enforced before any code is generated
Timeline slips and Phase 3 never ships	LOW	<i>Revenue features (fees, Paystack) delayed</i>	Phase 3 scope is stretch goal — contractually separate from launch

7.3 What Happens After September 7th

- Phase 3 (fees, Paystack, analytics, timetable) begins if timeline is beaten or in October
- Second school onboarding begins — use Child Prime Onyx as the demo reference
- Bug fixes and feedback from Child Prime Onyx incorporated in weeks 14–16
- Pricing formalised and subscription billing activated for new schools
- Chronix Technology Limited begins active sales to other Lagos private schools